

WAVELET BASIS FOR SOLVING PDEs

ME5204 FINITE ELEMENT ANALYSIS (COURSE PROJECT)

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Abstract

This project delves into the development of a multi-scale finite element framework utilizing hierarchical wavelet-like basis functions to effectively solve partial differential equations (PDEs) with complex, multi-scale features. The methodology incorporates the construction of the mass matrix, stiffness matrix, and forcing vector at the element level. The hierarchical basis functions are designed to capture both coarse- and fine-scale behavior. The stiffness matrix captures contributions from both coarse- and fine-scale basis functions with scale-dependent coefficients. The mass matrix and forcing vector are constructed to reflect the hierarchical nature of the basis and its interaction with the source function using numerical integration. This multi-scale framework, utilizing wavelet-like basis functions, offers a robust approach to solving PDEs

Keywords: multi-scale finite elements, wavelet basis, elemental matrices

1 Introduction

Many physical and engineering problems are governed by partial differential equations (PDEs) that describe processes such as heat conduction, fluid flow, and structural deformation. Solving these equations numerically often requires hierarchical methods to capture both coarse-scale and fine-scale features of the solution. Multi-scale finite element methods (MsFEM) have proven effective in addressing this challenge by leveraging wavelet-like basis functions. The use of hierarchical basis functions is particularly advantageous in problems requiring scale separation. These basis functions enable efficient representation of solutions across different scales, ensuring high accuracy and computational efficiency.

Figure 1 inspired from [2] illustrates a multilevel collection Ψ of functions defined on the interval $[0, 1]$. The index set $\vee_{\mathcal{J}}$ with parent-child relations is shown on the left, with a tree $\Lambda \subset \vee_{\mathcal{J}}$ visualized in red. This collection of functions is called the *hierarchical basis*.

The middle image in Figure 1 represents the hierarchical basis $\psi_{\lambda}(x)$, which is a multilevel collection of functions defined on the interval $[0, 1]$. Each function in the hierarchical basis is localized to a specific subinterval and supports the construction of nested subspaces:

$$V_k = V_{k-1} \oplus W_k, \quad \text{for } k = 1, 2, \dots,$$

where W_k is the wavelet complement space spanned by the hierarchical basis functions at level k . These functions satisfy:

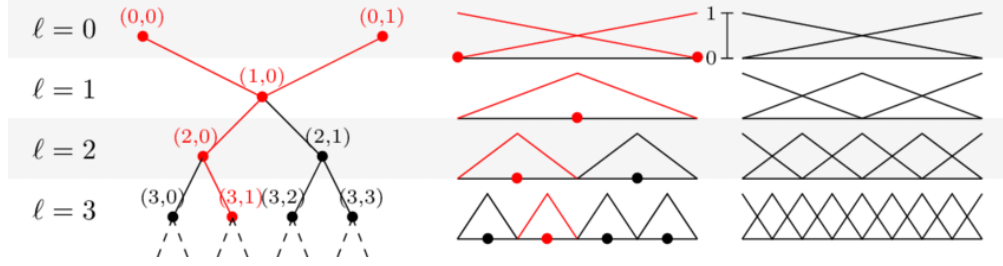


Figure 1: Index set $\mathcal{V}_{\mathcal{J}}$ and tree $\Lambda \subset \mathcal{V}_{\mathcal{J}}$. Multilevel functions Ψ , Scaling functions Φ

1. Compact support: $\psi_{\lambda}(x)$ is nonzero only on $S(\lambda) := \text{supp}(\psi_{\lambda})$, a subinterval of $[0, 1]$.
2. Refinement: Functions at finer levels refine the approximation by adding localized detail.
3. Partition of unity: The hierarchical basis satisfies $\sum_{\lambda \in \mathcal{V}_{\mathcal{J}}} \psi_{\lambda}(x) = 1$, ensuring completeness.

In contrast, the rightmost image illustrates the standard finite element basis $\phi_i(x)$, which forms a piecewise linear approximation over the domain $[0, 1]$. Each $\phi_i(x)$ is defined over two adjacent subintervals, and the collection of these functions spans a single subspace V_k without hierarchical refinement:

$$V_k = \text{span}\{\phi_1(x), \phi_2(x), \dots, \phi_n(x)\}.$$

Key properties of the finite element basis:

1. Support: Each basis function $\phi_i(x)$ is supported on exactly two elements.
2. Locality: The basis functions are strictly local, enabling efficient assembly of the stiffness and mass matrices.
3. Simplicity: The finite element basis uses a single resolution level, lacking the multiscale adaptability of $\psi_{\lambda}(x)$.

This report covers the formulation of the governing equations, their weak forms, and the discretization approach using hierarchical basis functions. The construction of stiffness and mass matrices and the forcing vector is detailed, followed by numerical examples validating the methodology. The report concludes with observations and potential applications.

2 Governing equations, weak form and the hierarchical Schauder Basis

We consider the boundary value problem:

$$-\epsilon \nabla^2 u + cu = f \quad \text{on } X,$$

with the following specifications:

- ϵ is a constant representing the *diffusion* or *conductivity coefficient*.
- c is a coefficient that models a *reaction* or *stiffness term*.
- u is the *unknown function* to be solved for, such as displacement, temperature, etc.

- f is the *source term* or *external forcing function*.

The domain of the problem is X , typically the interval $[0, 1]$, and the boundary conditions on u are homogeneous Dirichlet conditions:

$$u = 0 \quad \text{on } \partial X.$$

Weak Formulation

To solve the governing equation numerically, we transform it into a weak form. The weak form is derived by multiplying the equation by a test function $v \in H_0^1(X)$, the space of functions whose first derivatives are square-integrable and vanish on the boundary. We then integrate over the domain and apply integration by parts to reduce the order of the derivatives. This gives the weak form:

$$\int_X \epsilon \nabla u \cdot \nabla v \, dx + \int_X c u v \, dx = \int_X f v \, dx.$$

This formulation is advantageous as it allows for functions with lower regularity and handles boundary conditions in a more natural way.

We define the bilinear form:

$$a(u, v) = \int_X \epsilon \nabla u \cdot \nabla v \, dx + \int_X c u v \, dx,$$

and the linear form:

$$F(v) = \int_X f v \, dx.$$

This leads to the weak formulation of the problem:

$$a(u, v) = F(v),$$

Finite Element Approximation

We now approximate the solution u using a finite-dimensional subspace V_k at scale k . The weak form of the problem in this subspace is:

$$\int_X \epsilon \nabla u_k \cdot \nabla \phi_k \, dx + \int_X c u_k \phi_k \, dx = \int_X f \phi_k \, dx,$$

where $u_k \in H_0^1$, and ϕ_k is the finite-dimensional basis for V_k at scale k . In matrix form, the weak form becomes:

$$M_k u_k + K_k u_k = F_k,$$

where M_k is the mass matrix, K_k the stiffness matrix, and F_k the force vector. Alternatively, the bilinear form is written as:

$$a(\phi_k, \phi_k) = F(\phi_k).$$

Alternative Basis and Wavelet Transformation

According to [1], we now consider an alternative basis for V_k defined as:

$$\psi_k = W_k^T \phi_k,$$

where W_k is a nonsingular $N \times N$ matrix. Using this transformation, u_k can be expressed as:

$$u_k = \Delta u^T \psi_k,$$

which also may be found by solving the equation:

$$a(\psi_k, \psi_k) \Delta u = F(\psi_k).$$

This results in the preconditioned linear system:

$$A_k^\psi = W_k^T A_k^\phi W_k,$$

where $A_k^\phi = M_k + K_k$. The subspaces V_k satisfy:

$$V_0 \subset V_1 \subset \cdots \subset V_k, \quad \bigcup_k V_k = H.$$

For $k \geq 1$, the Schauder basis satisfies:

$$V_k = V_{k-1} \oplus W_k.$$

Wavelet Transform and Two-Scale Transformation

The wavelet transform can be written recursively as:

$$W_k = \begin{bmatrix} T_k & 0 \\ 0 & I_{k-1} \end{bmatrix} W_{k-1},$$

One choice for W_k is the hierarchical Schauder basis. This basis consists of the elements of the linear finite element basis ϕ at multiple scales. The relationship between ψ_k and ϕ_k is simply:

$$\psi_k^j = \phi_k^{2^{j-1}}, \quad j = 1, \dots, N_{\text{np}}^k,$$

where ϕ_k is the linear finite element basis. Here, the superscript j indicates the node number, k indicates the scale, and N_{np}^k is the number of node points at scale k .

The Schauder basis is illustrated in Figure 2. the construction of the two-scale transformation can be written in terms of the grid spacing. To define the wavelet transform, let:

- H_k be the $(N_{\text{np}}^k - 1) \times (N_{\text{np}}^{k-1} - 1)$ matrix, where $H_k = [h_k^{j,j'}]$, and
- G_k be the $(N_{\text{np}}^k - 1) \times (N_{\text{np}}^{k-1} - 1)$ matrix, where $G_k = [g_k^{j,j'}]$. where T_k is the two-scale transformation matrix.

$$T_k = [H_k | G_k] \tag{13}$$

The function values $h_k^{jj'}$ and $g_k^{jj'}$ for ϕ_k^j and ψ_k^j respectively at the node points x_k^j are given by

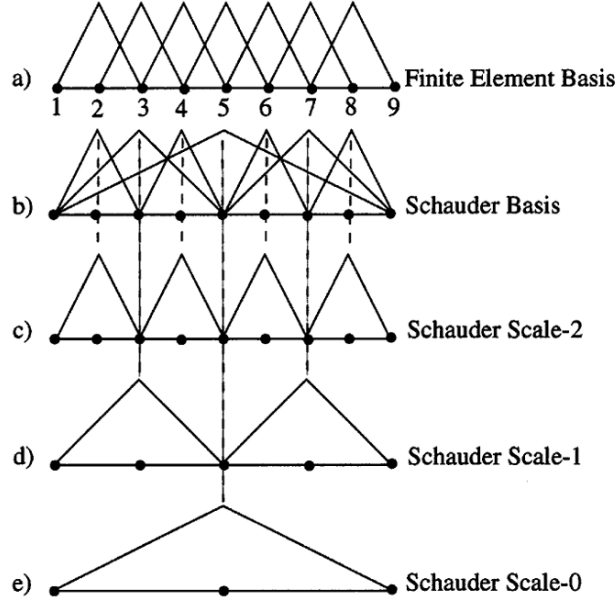


Figure 2: The hierarchical Schauder basis for scale $k = 2$ with 8 elements and 9 nodes showing (a) the equivalent finite element basis, (b) the composite basis, (c) the basis functions for Scale-2, (d) the basis functions for Scale-1, and (e) the basis functions for Scale-0. [1]

$$h_k^{jj'} = \begin{cases} \frac{\Delta_k^j}{\Delta_{k-1}^{j'}}, & j' = 2j - 1, \\ 1, & j' = 2j, \\ \frac{\Delta_k^{j'+1}}{\Delta_{k-1}^{j'}}, & j' = 2j + 1, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

$$g_k^{jj'} = \delta_{j', 2j-1}. \quad (15)$$

In one dimension, the element length from x_k^{j-1} to x_k^j is defined as

$$\Delta_k^j = x_k^j - x_k^{j-1}.$$

$$\phi_{k-1}^j = \sum h_k^{j,j'} \phi_k^{j'}, \quad \psi_k^j = \sum g_k^{j,j'} \phi_k^{j'}.$$

The Schauder basis provides compact support, enabling efficient multi-scale analysis.

3 Discretization approach

Hierarchical wavelet basis functions provide a powerful approach for refining the solution within an element. The main idea is to capture the coarse-scale behavior of the solution using finite element basis functions ϕ_i and then refine it further by adding contributions from wavelet basis functions $\psi_{l,j}$ that operate at different hierarchical scales. This hierarchical refinement ensures that both global and local solution features are accurately represented.

For each element, the solution is expressed as a combination of a coarse-scale contribution and a series of fine-scale corrections:

$$\nu(x) = \sum_i C_i \phi_i(x) + \sum_{l=0}^{k-1} \sum_{j=0}^{2^l-1} C_{l,j} \psi_{l,j}(x), \quad (1)$$

where:

- $\phi_i(x)$ are the finite element basis functions, capturing the coarse solution.
- $\psi_{l,j}(x)$ are the hierarchical wavelet basis functions at hierarchical level l and local shift j , capturing fine-scale details within the element.
- C_i are the coefficients for the finite element basis functions.
- $C_{l,j}$ are the coefficients for the wavelet basis functions.
- k is the total number of hierarchical levels.
- l represents the current hierarchical level, ranging from 0 (coarsest level) to $k-1$ (finest level).
- j is the index of the wavelet basis function within a specific hierarchical level l , ranging from 0 to $2^l - 1$.

The hierarchical structure ensures that the coarse solution is computed first using ϕ_i , and then finer details are progressively added through the wavelet coefficients $C_{l,j}$. This allows the method to adaptively refine the solution where needed, without globally refining the mesh. The wavelet basis functions $\psi_{l,j}(x)$ have local support within the element, making them ideal for capturing high-frequency details.

This section describes the detailed derivation of the mass, stiffness, and forcing matrices for the hierarchical wavelet basis. It also outlines the procedure for reconstructing the coefficients and interpolating the solution.

3.1 Hierarchical Basis Functions in $\psi_{l,j}$ Notation

The hierarchical wavelet basis functions are defined as $\psi_{l,j}(\xi)$, where l is the hierarchical level, and j is the index of the wavelet within that level. Below, we list the functions for levels 0 through 3 in the parametric domain $\xi \in [-1, 1]$.

Level 0 (Coarse Level)

The coarse-level (global) shape functions are:

$$\phi_1(\xi) = \frac{1-\xi}{2}, \quad \phi_2(\xi) = \frac{1+\xi}{2}.$$

Level 1 (Wavelet Level 1)

At level 1, a single wavelet is introduced:

$$\psi_{1,1}(\xi) = 1 - |\xi|.$$

Level 2 (Wavelet Level 2)

At level 2, two hierarchical wavelets are introduced, each supported over half the domain:

$$\psi_{2,1}(\xi) = \begin{cases} 2(1 + \xi), & \xi \in [-1, -\frac{1}{2}], \\ 2(1 - \xi), & \xi \in [-\frac{1}{2}, 0], \\ 0, & \text{otherwise,} \end{cases}$$
$$\psi_{2,2}(\xi) = \begin{cases} 0, & \xi \in [-1, 0], \\ 2(1 + \xi), & \xi \in [0, \frac{1}{2}], \\ 2(1 - \xi), & \xi \in [\frac{1}{2}, 1], \\ 0, & \text{otherwise.} \end{cases}$$

Level 3 (Wavelet Level 3)

At level 3, four hierarchical wavelets are introduced, each supported over a quarter of the domain:

$$\psi_{3,1}(\xi) = \begin{cases} 4(1 + \xi), & \xi \in [-1, -\frac{3}{4}], \\ 4(1 - \xi), & \xi \in [-\frac{3}{4}, -\frac{1}{2}], \\ 0, & \text{otherwise,} \end{cases}$$
$$\psi_{3,2}(\xi) = \begin{cases} 4(1 + \xi), & \xi \in [-\frac{1}{2}, -\frac{1}{4}], \\ 4(1 - \xi), & \xi \in [-\frac{1}{4}, 0], \\ 0, & \text{otherwise,} \end{cases}$$
$$\psi_{3,3}(\xi) = \begin{cases} 0, & \xi \in [-1, 0], \\ 4(1 + \xi), & \xi \in [0, \frac{1}{4}], \\ 4(1 - \xi), & \xi \in [\frac{1}{4}, \frac{1}{2}], \\ 0, & \text{otherwise,} \end{cases}$$
$$\psi_{3,4}(\xi) = \begin{cases} 0, & \xi \in [-1, \frac{1}{2}], \\ 4(1 + \xi), & \xi \in [\frac{1}{2}, \frac{3}{4}], \\ 4(1 - \xi), & \xi \in [\frac{3}{4}, 1], \\ 0, & \text{otherwise.} \end{cases}$$

3.2 General Characteristics of $\psi_{l,j}$ Functions

- At level $l = 0$, $\phi_1(\xi)$ and $\phi_2(\xi)$ are the linear shape functions spanning the entire domain.
- For $l > 0$, the $\psi_{l,j}(\xi)$ functions are piecewise linear wavelet functions, with compact support that becomes more localized with increasing l .
- The index j determines the position of the wavelet within the domain at the given level.

3.3 Mass Matrix

The mass matrix for a single element is defined as:

$$M_{ij} = \int_{-1}^1 \phi_i(\xi) \phi_j(\xi) J d\xi, \quad (2)$$

where $\phi_i(\xi)$ and $\phi_j(\xi)$ are the hierarchical wavelet basis functions, and J is the Jacobian of the transformation:

$$J = \frac{h}{2}, \quad h = \text{element length}. \quad (3)$$

The mass matrix consists of three types of contributions:

- **Coarse-Coarse:** Contributions from finite element basis functions ϕ_i and ϕ_j .
- **Coarse-Fine:** Interactions between finite element basis functions ϕ_i and wavelet basis functions $\psi_{l,j}$.
- **Fine-Fine:** Interactions between wavelet basis functions ψ_{l_1,j_1} and ψ_{l_2,j_2} .

For coarse-scale contributions:

$$M_{00} = \int_{-1}^1 \phi_0(\xi) \phi_0(\xi) J d\xi, \quad (4)$$

$$M_{01} = \int_{-1}^1 \phi_0(\xi) \phi_1(\xi) J d\xi, \quad (5)$$

$$M_{11} = \int_{-1}^1 \phi_1(\xi) \phi_1(\xi) J d\xi. \quad (6)$$

For coarse-fine contributions:

$$M_{0,l,j} = \int_{-1}^1 \phi_0(\xi) \psi_{l,j}(\xi) J d\xi, \quad (7)$$

$$M_{1,l,j} = \int_{-1}^1 \phi_1(\xi) \psi_{l,j}(\xi) J d\xi. \quad (8)$$

For fine-fine contributions:

$$M_{(l_1,j_1),(l_2,j_2)} = \int_{-1}^1 \psi_{l_1,j_1}(\xi) \psi_{l_2,j_2}(\xi) J d\xi. \quad (9)$$

The structure of the mass matrix is illustrated in Figure 3, where non-zero elements correspond to the interactions described above. Coarse-coarse contributions populate the upper-left block, coarse-fine contributions populate off-diagonal blocks, and fine-fine contributions populate diagonal and off-diagonal blocks at the finer scales.

3.3.1 Quadrature Rule

To evaluate the integrals, we use a 1D Gaussian quadrature rule with 3 points:

- Quadrature points: $\xi_1 = -\sqrt{\frac{3}{5}}, \xi_2 = 0, \xi_3 = \sqrt{\frac{3}{5}}$.
- Weights: $w_1 = \frac{5}{9}, w_2 = \frac{8}{9}, w_3 = \frac{5}{9}$.

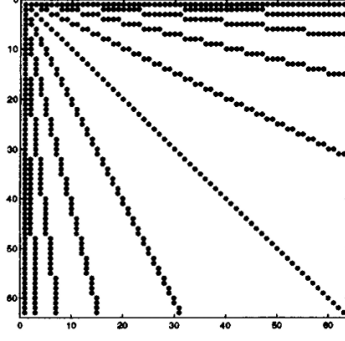


Figure 3: Non-zero pattern of the hierarchical mass matrix [1]

The integral is approximated as:

$$\int_{-1}^1 f(\xi) d\xi \approx \sum_{i=1}^3 w_i f(\xi_i). \quad (10)$$

This rule ensures accurate integration of polynomials up to degree 5, which is sufficient for the hierarchical basis functions.

3.4 Stiffness Matrix

Hierarchical basis functions, as illustrated in Figure 4, play a critical role in capturing both coarse and fine solution features. Each level of the hierarchy introduces finer basis functions that contribute to the overall representation of the solution. Their derivatives are essential for constructing the stiffness matrix, which represents the weak form of the governing equations.

3.4.1 Definition of Derivatives

The derivatives of the hierarchical basis functions are defined to ensure continuity and refinement at each hierarchical level. For a given scale, the derivatives capture the gradient of the solution refinement introduced at that level:

$$\frac{d\phi_i}{dx}, \quad \frac{d\psi_{l,j}}{dx}.$$

Figure 4 demonstrates how the hierarchical basis functions and their corresponding derivatives are structured. For each hierarchical level:

- **Coarse-scale derivatives:** The gradients of $\phi_i(x)$ represent the primary variation of the solution.
- **Fine-scale derivatives:** The gradients of $\psi_{l,j}(x)$ capture high-frequency variations.

3.4.2 Contribution to the Stiffness Matrix

The stiffness matrix is constructed using the derivatives of the hierarchical basis functions:

$$K_{ij} = \int_{-1}^1 \frac{d\phi_i(\xi)}{d\xi} \frac{d\phi_j(\xi)}{d\xi} J d\xi, \quad (11)$$

where $J = \frac{h}{2}$ is the Jacobian of the transformation from the reference to the physical element. The stiffness matrix contains three types of contributions:

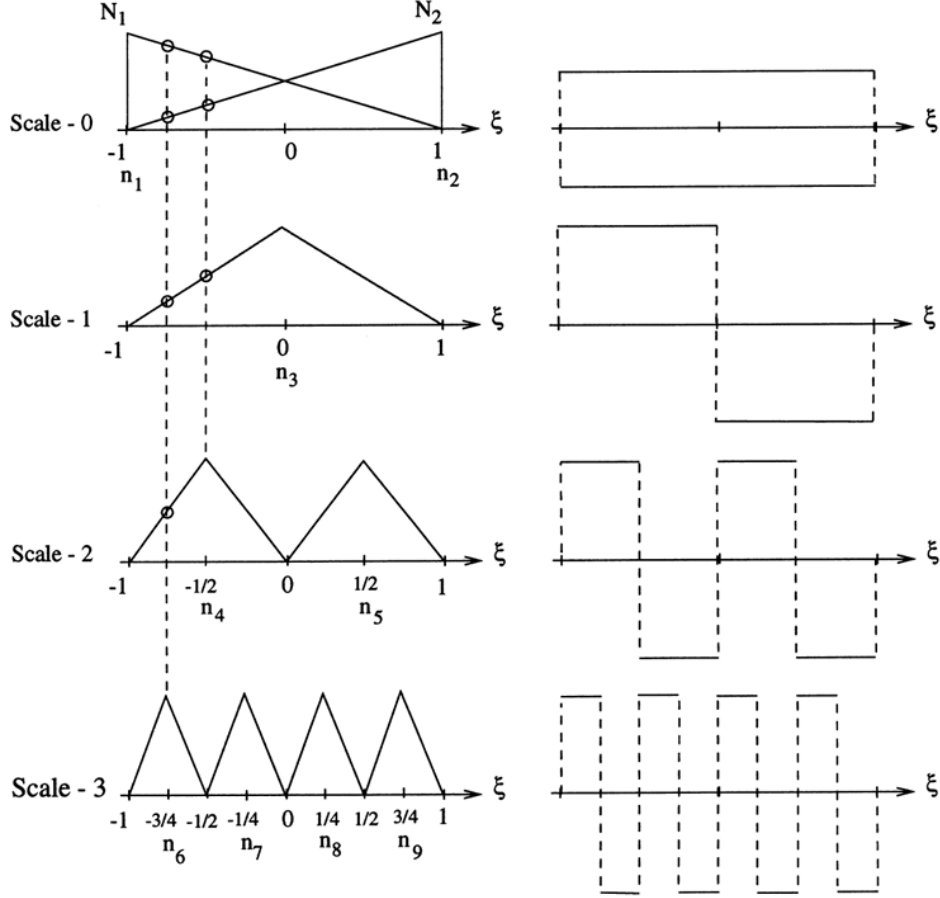


Figure 4: Hierarchical basis functions (left) and their derivatives (right) at various scales. The refinement from coarse to fine scales is evident in both basis functions and their derivatives. [1]

- **Coarse-Coarse:** Interactions between derivatives of finite element basis functions, ϕ_i .
- **Coarse-Fine:** Interactions between derivatives of finite element basis functions, ϕ_i , and wavelet basis functions, $\psi_{l,j}$.
- **Fine-Fine:** Interactions between derivatives of wavelet basis functions, ψ_{l_1,j_1} and ψ_{l_2,j_2} .

For example:

$$K_{00} = \int_{-1}^1 \frac{d\phi_0}{d\xi} \frac{d\phi_0}{d\xi} J d\xi, \quad (12)$$

$$K_{0,l,j} = \int_{-1}^1 \frac{d\phi_0}{d\xi} \frac{d\psi_{l,j}}{d\xi} J d\xi, \quad (13)$$

$$K_{(l_1,j_1),(l_2,j_2)} = \int_{-1}^1 \frac{d\psi_{l_1,j_1}}{d\xi} \frac{d\psi_{l_2,j_2}}{d\xi} J d\xi. \quad (14)$$

The stiffness matrix for a single element is defined as:

$$K_{ij} = \int_{-1}^1 \frac{d\phi_i}{d\xi} \frac{d\phi_j}{d\xi} \frac{1}{J} d\xi, \quad (15)$$

where $\frac{d\phi}{d\xi}$ are the derivatives of the hierarchical basis functions.

For coarse-scale contributions:

$$K_{00} = \int_{-1}^1 \frac{d\phi_0}{d\xi} \frac{d\phi_0}{d\xi} \frac{1}{J} d\xi, \quad (16)$$

$$K_{01} = \int_{-1}^1 \frac{d\phi_0}{d\xi} \frac{d\phi_1}{d\xi} \frac{1}{J} d\xi, \quad (17)$$

$$K_{11} = \int_{-1}^1 \frac{d\phi_1}{d\xi} \frac{d\phi_1}{d\xi} \frac{1}{J} d\xi. \quad (18)$$

For fine-scale contributions:

$$K_{ij} = \int_{-1}^1 \frac{d\psi_{l_1,j_1}}{d\xi} \frac{d\psi_{l_2,j_2}}{d\xi} \frac{1}{J} d\xi. \quad (19)$$

The stiffness matrix is highly sparse because contributions exist only in the first 2×2 block (coarse-scale) and along the diagonal (fine-scale). All off-diagonal fine-fine entries are zero due to the orthogonality of hierarchical basis functions. This sparsity structure ensures computational efficiency and aligns with the reference pattern illustrated in Figure 5.

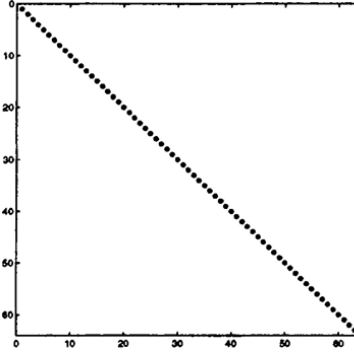


Figure 5: Non-zero pattern of the hierarchical stiffness matrix. [1]

3.4.3 Quadrature Rule

Numerical integration for constructing the stiffness matrix employs the Gauss-Legendre quadrature rule. The quadrature weights w_q and points ξ_q are chosen to exactly integrate polynomials of degree $2n - 1$ or less, where n is the number of quadrature points. The integral is approximated as:

$$\int_{-1}^1 f(\xi) d\xi \approx \sum_{q=1}^n w_q f(\xi_q). \quad (20)$$

The choice of quadrature points depends on the polynomial degree of the basis functions. For hierarchical wavelet basis functions, the number of quadrature points must be sufficient to capture both coarse-scale and fine-scale contributions accurately.

3.5 Forcing Vector

The forcing vector is defined as:

$$F_i = \int_{-1}^1 f(\xi) \phi_i(\xi) J d\xi, \quad (21)$$

where $f(\xi)$ is the source function. The contributions to the forcing vector are computed for both coarse-scale and fine-scale basis functions:

$$F_0 = \int_{-1}^1 f(\xi) \phi_0(\xi) J d\xi, \quad (22)$$

$$F_1 = \int_{-1}^1 f(\xi) \phi_1(\xi) J d\xi, \quad (23)$$

$$F_{ij} = \int_{-1}^1 f(\xi) \psi_{i,j}(\xi) J d\xi. \quad (24)$$

Numerical integration with Gaussian quadrature, as discussed for the stiffness and mass matrix is used to compute these integrals.

3.6 Reconstruction of Coefficients

The coefficients for the hierarchical wavelet basis are computed element by element. For each element, the governing equation is given by:

$$(K_{uu} + M_{uu})C_u = F_u - (K_{ub} + M_{ub})C_b, \quad (25)$$

where:

- K and M are the stiffness and mass matrices, respectively, for the element.
- F_u is the forcing vector for the element corresponding to the unknown indices.
- C_u represents the unknown coefficients at the interior nodes.
- C_b are the coefficients at boundary nodes, which are known (e.g., from boundary conditions or continuity requirements).
- K_{ub} and M_{ub} are submatrices corresponding to interactions between interior and boundary nodes.

The primary task is to solve for the unknown coefficients C_u at the interior nodes. The equation can be rearranged as:

$$C_u = (K_{uu} + M_{uu})^{-1} (F_u - (K_{ub} + M_{ub})C_b), \quad (26)$$

where K_{uu} and M_{uu} are the stiffness and mass submatrices for the interior nodes, and F_u is the forcing vector associated with the interior nodes.

3.6.1 Steps for Computing Coefficients

The computation of coefficients involves the following steps:

1. **Determine Known Values:** For each element, identify the nodes at the element boundaries (e.g., n_1, n_2 in the Figure 4). The coefficients at these nodes, C_b , are known by solving the traditional FEM at these nodes
2. **Extract Submatrices:** Partition the global stiffness matrix K and mass matrix M into submatrices K_{uu} , K_{ub} , M_{uu} , and M_{ub} . These submatrices represent the interactions between interior nodes and boundary nodes, as well as within the interior nodes themselves.

3. **Compute Forcing Contributions:** Calculate the interior forcing vector F_u for the element. Subtract the contributions of the boundary terms, $(K_{ub} + M_{ub})C_b$, to isolate the contributions of the unknowns.
4. **Solve for Interior Coefficients:** Solve the linear system for C_u :

$$C_u = (K_{uu} + M_{uu})^{-1} (F_u - (K_{ub} + M_{ub})C_b). \quad (27)$$

This provides the coefficients at the interior nodes of the element.

5. **Assemble Global Coefficients:** Once the interior coefficients C_u are computed for all elements, they are assembled into the global coefficient vector C . The known boundary coefficients C_b are directly inserted at the corresponding positions in the global vector.

3.7 Interpolation of the Solution

For hierarchical wavelets, the interpolation includes contributions from both coarse-scale and fine-scale basis functions:

$$u(x) = \sum_{i=0}^1 C_i \phi_i(x) + \sum_{l=0}^{k-1} \sum_{j=0}^{2^l-1} C_{l,j} \psi_{l,j}(x). \quad (28)$$

The interpolation is performed element-wise, mapping physical coordinates to the reference element:

$$\xi = \frac{2x - (x_1 + x_2)}{x_2 - x_1}. \quad (29)$$

The reconstructed solution is then computed by evaluating the basis functions at the desired points.

4 Numerical Example

4.1 Example 1

To validate the hierarchical wavelet basis method, we start with a simple numerical example. The governing equation is given as:

$$-\epsilon \nabla^2 u + cu = f(x), \quad x \in [0, 1], \quad (30)$$

where:

- $\epsilon = 1$: Diffusion coefficient.
- $c = 0$: Convection coefficient.
- $f(x) = 1$: Source term.

The boundary conditions for this example are:

$$u(0) = 0, \quad u(1) = 0. \quad (31)$$

We assume that only 2 elements are present and compare the exact solution and the solutions obtained from scale 0,1,2.

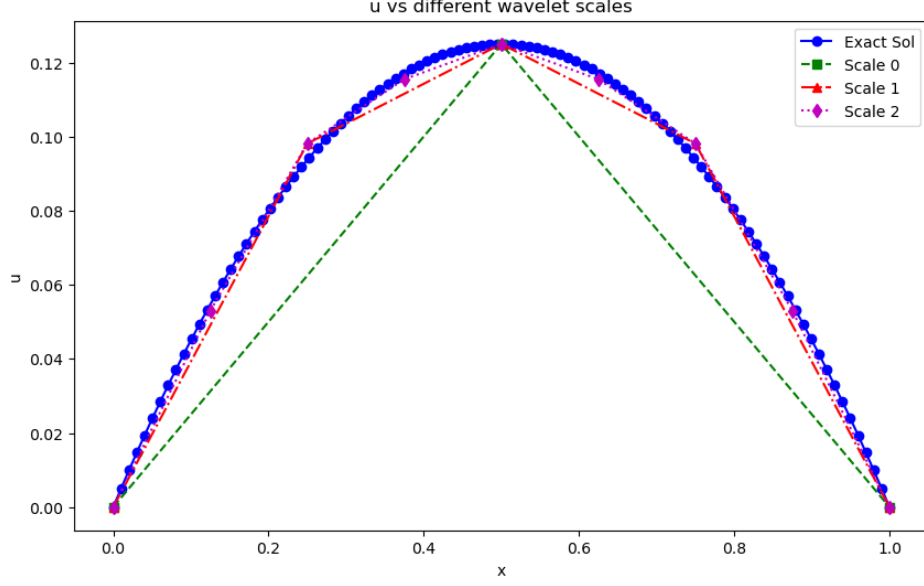


Figure 6: Solution vs wavelet scaling

This shows that the solution is getting closer to the exact solution just by increasing the number of wavelets used inside each element

4.2 Convergence

Since obtaining convergence in the finite element space is straightforward, we will focus on the wavelet scale convergence.

Wavelet methods decompose the solution space into different scales, with each scale capturing finer details of the solution. The optimal value of the scale refers to the point at which increasing the resolution further (i.e., going to finer scales) no longer significantly improves accuracy or the computational cost outweighs the accuracy gained.

Here, we explore the wavelet scale convergence for Example 1 as given in the previous section

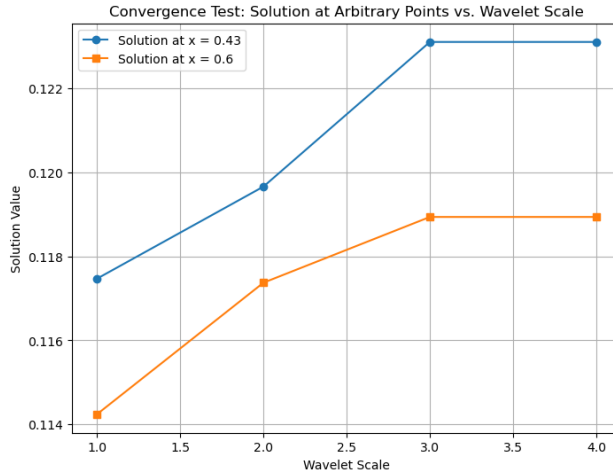


Figure 7: Wavelet scale vs solution value at $x = 0.43$ and $x = 0.6$

x coordinate	solution value (u)
$x = 0.43$	0.107
$x = 0.6$	0.1

Table 1: Finite Element Solution (Scale 0) with 2 elements

What we can infer from this is, going beyond wavelet scale 3 is not advisable as there is minimum change in the value of the solution

The values in Table 1 correspond to the solution value purely through Finite Element Basis function. We can see there is significant difference between them. Finite Element basis being less accurate.

4.3 FEM vs Wavelet Basis

To give a final comparison between the hierarchical wavelet basis method and finite element basis method we study this example. The governing equation is given as:

$$-\epsilon \nabla^2 u + cu = f(x), \quad x \in [0, 1], \quad (32)$$

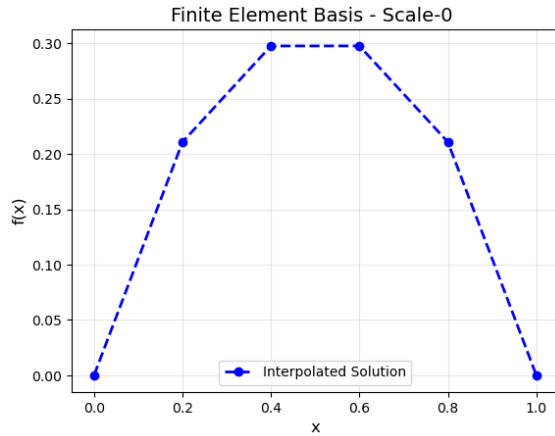
where:

- $\epsilon = 10$: Diffusion coefficient.
- $c = 100$: Convection coefficient.
- $f(x) = 50$: Source term.

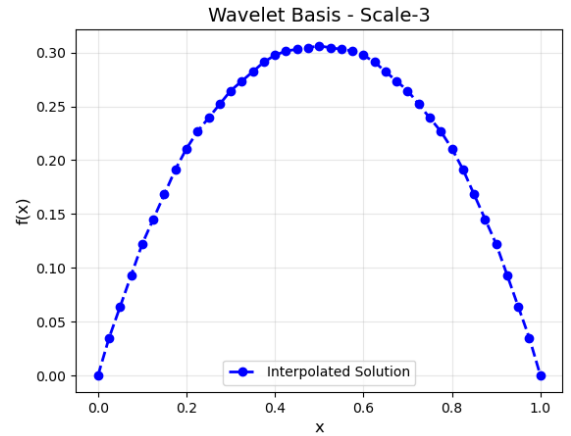
The boundary conditions for this example are:

$$u(0) = 0, \quad u(1) = 0. \quad (33)$$

We assume total elements = 5 and compare the solutions obtained from FEM (scale 0) and Wavelet Basis scale 3.



(a) Solution obtained using FEM Basis Functions (Scale 0)



(b) Solution obtained using Wavelet Basis Functions (Scale 3)

Figure 8: Comparison of FEM and Wavelet Basis Function Solutions.

This clearly shows that the hierarchical basis function works much better than the finite element basis when the number of elements are a small quantity

Thus, this approach is more efficient than normal FEM, giving better results for the same number of elements

5 Conclusions

- The hierarchical wavelet basis outperforms the finite element basis, especially for a small number of elements, due to its ability to capture both coarse and fine details efficiently.
- Wavelet-based methods provide higher accuracy than traditional finite element methods for the same number of elements, with reduced numerical error.
- Convergence analysis shows that increasing the wavelet scale improves accuracy, particularly for high-frequency details. Scale 3 is identified as optimal, balancing accuracy and computational cost.
- The wavelet basis effectively handles multi-scale features by separating coarse-scale behavior from fine-scale corrections, avoiding global mesh refinement.
- Sparse mass and stiffness matrices in the wavelet framework ensure computational efficiency while maintaining solution accuracy.
- Numerical results closely match exact solutions, validating the wavelet-based implementation's robustness for constant and non-constant forcing terms.
- Compared to finite element methods, wavelet basis solutions resolve sharp gradients and localized features better, as demonstrated in tests with $\epsilon = 10$, $c = 100$, and $f(x) = 50$.
- This approach is ideal for problems requiring localized refinement, such as heat conduction, fluid flow, and structural deformation with complex geometries.
- The hierarchical nature of the wavelet basis enables adaptive refinement, making it scalable for time-dependent and higher-dimensional problems.

The project successfully demonstrated the utility of hierarchical wavelet basis functions for solving PDEs. The combination of coarse and fine basis functions ensures a balance between computational efficiency and accuracy. The ability to control the refinement through wavelet levels makes this method adaptable and robust for a wide range of problems. Future extensions could explore adaptive wavelet refinement, multi-dimensional applications, and integration with time-dependent problems.

References

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